

A Strategic Decision Support System for Customer Churn Prediction

Phase 1 Project Proposal

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Course: Strategic Information Systems

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Abstract

Customer churn is one of the most strategically consequential problems in the telecommunications industry because it directly affects revenue stability, customer lifetime value, and retention expenditure. Within Strategic Information Systems, churn analytics should not be treated as a stand-alone classification exercise; rather, it should function as a decision-support capability that helps managers identify risk, interpret its drivers, and select timely interventions. This Phase 1 project proposal presents a strategic decision support system centered on customer churn prediction using the public IBM Telco Customer Churn dataset hosted on Kaggle. The proposed study combines data preprocessing, feature engineering, class-imbalance handling, comparative machine learning modeling, and explainable AI techniques to produce both predictive and managerial outputs. The reviewed literature indicates that tabular ensemble methods remain highly effective for telecom churn data, while recent research increasingly emphasizes transparency, business alignment, and reproducible evaluation. Building on these insights, the proposed methodology will compare interpretable and high-performing models, evaluate them with imbalance-aware metrics, and translate model outputs into risk tiers and action recommendations for retention management. The intended contribution is an interpretable, business-oriented analytics artifact that supports analytical CRM and strategic resource allocation rather than merely producing a binary churn label.

Keywords: *customer churn prediction, telecommunications, decision support system, analytical CRM, explainable AI, retention strategy*

1. Introduction

In saturated telecommunications markets, service bundles are increasingly imitable and switching costs are often low, which makes customer retention a core competitive capability rather than a purely operational concern. Recent telecom churn studies reiterate that retaining an existing subscriber is substantially less expensive than acquiring a new one and that even moderate improvements in churn detection can materially influence profitability (Chang et al., 2024; Sikri et al., 2024; El Attar & El-Hajj, 2026). For this reason, churn analytics is directly relevant to strategic planning, revenue protection, and customer relationship management.

From a Strategic Information Systems perspective, the value of churn analytics lies not only in predictive accuracy but also in whether the system helps managers allocate retention budgets, personalize interventions, interpret risk factors, and coordinate marketing, service, and pricing decisions. A useful system must therefore move beyond a simple binary classifier. It should operate as an analytical decision support system (DSS) that converts model outputs into prioritized customer lists, explanatory drivers, and recommended response strategies.

This proposal addresses that need through a project titled A Strategic Decision Support System for Customer Churn Prediction. The study will use the public IBM Telco Customer Churn benchmark from Kaggle (Yeanzc, n.d.), which is widely used in contemporary churn research because it contains customer demographics, contract characteristics, service subscriptions, and billing information in a structured tabular format. The dataset is especially suitable for a Phase 1 proposal because it allows rigorous and reproducible planning while remaining closely aligned with real analytical CRM questions in the telecom sector.

The main objective is to design an interpretable churn analytics pipeline that can support managerial decision making in customer retention. More specifically, the project aims to compare multiple machine learning models, identify the main drivers of churn, and embed the best-performing predictive core in a recommendation-oriented DSS. Accordingly, the study is guided by the following research questions:

- 1. RQ1. Which customer attributes and tabular machine learning models provide the most reliable churn predictions on the IBM Telco dataset?
- 2. RQ2. How can explainable AI techniques translate model outputs into interpretable reasons for churn risk?
- 3. RQ3. How can churn predictions be operationalized within a strategic DSS that supports retention prioritization and action planning?

The remainder of this Phase 1 report reviews recent literature on churn modeling and then presents the proposed materials and methods for the final implementation stage.

2. Literature Review

Recent churn prediction research has evolved from classical classification approaches toward a broader landscape that includes ensemble learning, deep learning, explainable AI, profit-aware evaluation, and adaptive learning. The most comprehensive overview in the attached literature is provided by Imani et al. (2025), whose systematic review of 240 studies shows that boosting-based ensembles remain dominant on structured churn data, whereas deep learning is increasingly adopted for sequential and unstructured inputs. At the same time, the review highlights recurring limitations that are especially relevant for a Strategic Information Systems project: class imbalance, limited interpretability, concept drift, and the weak adoption of business-oriented evaluation criteria. Tables 1 and 2 synthesize the attached studies and their implications for the proposed project.

Study	Main focus	Representative findings	Implication for this project
Imani et al. (2025)	Systematic review of recent churn prediction research	Shows the dominance of ensemble ML on structured data and identifies interpretability, concept drift, and business-aligned evaluation as unresolved challenges.	Provides the macro-level agenda and gap structure for the proposal.

Study	Main focus	Representative findings	Implication for this project
Wagh et al. (2024)	Telecom churn modeling with up-sampling, ENN, random forest, and survival analysis	Reports substantial performance gains after balancing and explicitly links churn prediction to retention plans and survival-oriented thinking.	Justifies leakage-safe imbalance handling and motivates limited time-aware analysis through customer tenure.
Chang et al. (2024)	Ensemble learning and CRM-oriented explainability in telecom	Finds Random Forest strongest among evaluated models and combines prediction with LIME and SHAP to support CRM interpretation.	Supports the analytical CRM and explainability orientation of the DSS.
Sikri et al. (2024)	Ratio-based data balancing in telecom churn prediction	Demonstrates that the balancing strategy itself changes model behavior and that boosting methods benefit from better class representation.	Confirms that preprocessing is a strategic design choice, not a minor technical detail.
Omari et al. (2025)	IBM Telco benchmark with business KPI orientation	Reports strong SVM performance and frames churn analytics in relation to CLTV, ARPU, CAC, CRR, NPS, ethics, and transparency.	Reinforces the need for business-aligned and responsible analytics.

Table 1. Predictive modeling and data-preparation themes in the reviewed literature.

Study	Main focus	Representative findings	Implication for this project
Poudel et al. (2024)	Tabular models plus SHAP-based explanation	GBM outperforms the compared models in their setting and SHAP identifies contract type, tenure, and monthly charges as key churn drivers.	Demonstrates that prediction and explanation should be designed together.
Özkurt (2025)	Comparative evaluation of XAI methods on IBM Telco	Shows that SHAP is stronger for case-level explanations, while partial dependence offers useful population-level patterns.	Supports a multi-level interpretability strategy in the DSS.
Asif et al. (2025)	Explainable soft-voting ensemble on a large-scale telecom dataset	Combines XGBoost, CatBoost, LightGBM, SMOTE, Boruta, and SHAP/LIME to pursue both performance and transparency.	Illustrates a contemporary ensemble-plus-XAI architecture.
Hooda et al. (2026)	XCL-Churn explainable ensemble framework	Uses soft voting, advanced preprocessing, feature selection, and SHAP/LIME to combine predictive accuracy and interpretability.	Shows the current movement toward integrated, deployment-oriented churn systems.
Zerine et al. (2026)	Kaggle telecom benchmark with nested CV and model-agnostic explanations	Emphasizes PR-AUC, Wilcoxon tests, calibration-aware thinking, and analyst-friendly SHAP/PDP/ICE explanations.	Guides a more rigorous evaluation protocol for Phase 2.
El Attar & El-Hajj (2026)	Integrated framework with ensemble learning, SHAP, segmentation, and cost-sensitive thresholds	Bridges performance, explainability, segmentation, and business threshold selection to generate strategic risk profiles and recommended actions.	Most directly aligns with the decision-support goal of this project.

Table 2. Explainability, deployment, and decision-support themes in the reviewed literature.

2.1 Predictive modeling, class imbalance, and model choice

The literature consistently shows that model selection on telecom churn data is shaped by the structured, tabular nature of the features. Imani et al. (2025) identify boosting-based ensembles and tree methods as the dominant paradigm for structured churn prediction.

Telecom-specific evidence broadly supports this view. Wagh et al. (2024) report that random forests outperform decision trees after up-sampling and ENN, while also combining prediction with survival-oriented reasoning and retention plans. Chang et al. (2024) likewise demonstrate the strength of ensemble learning in telecom churn analysis, reporting superior performance for Random Forest relative to the other evaluated models.

Yet the literature also shows that predictive performance is sensitive to data preparation rather than model choice alone. Sikri et al. (2024) explicitly focus on imbalance handling and show that a ratio-based balancing technique can improve downstream predictive performance, with XGBoost and Gradient Boosting benefiting most. Omari et al. (2025), working on the IBM Telco dataset, find that an RBF-kernel SVM outperforms Logistic Regression, KNN, and Naive Bayes, reminding us that no single model family is universally superior and that benchmark comparisons should remain dataset-aware. Taken together, these studies suggest that the proposed project should compare both interpretable baselines and high-performing nonlinear learners while treating class imbalance and preprocessing as first-order design choices.

2.2 Explainability and managerial interpretability

A second major theme is the shift from pure prediction toward interpretable churn analytics. Poudel et al. (2024) argue that effective churn modeling has two inseparable components: predicting whether a customer will churn and explaining why. Their GBM-based study uses SHAP to generate both global and local explanations, showing that month-to-month contracts, tenure, and monthly charges are especially influential. Özkurt (2025) deepens this line of research by comparing SHAP with Interpreted ML Partial Dependence and concludes that the two techniques serve different analytical layers: SHAP is more informative for case-specific interpretation, whereas partial dependence clarifies aggregate feature effects.

Chang et al. (2024) similarly use LIME and SHAP to make ensemble predictions transparent enough for CRM-oriented action. This is strategically important because DSS users do not only need a risk score; they need a rationale that can be communicated to customer retention teams, marketing managers, and service operations. For this reason, explainability in the proposed project will be treated as a design requirement rather than a post hoc diagnostic.

2.3 Advanced ensembles, business alignment, and deployment-oriented evaluation

Recent studies increasingly attempt to combine strong predictive performance with deployable explanation layers. Asif et al. (2025) develop XAI-Churn TriBoost, a soft-voting ensemble of XGBoost, CatBoost, and LightGBM coupled with Bayesian imputation, multi-stage scaling, Boruta-based feature selection, SMOTE, and SHAP/LIME explanations. Hooda et al. (2026) propose a closely related XCL-Churn architecture that again unites gradient boosting, soft voting, feature selection, and explainability. Although these studies are based on larger telecom datasets than the IBM benchmark, they are methodologically valuable because they show that accuracy and interpretability can be pursued together instead of being framed as mutually exclusive objectives.

The literature also becomes more rigorous in evaluation. Zerine et al. (2026) benchmark five tabular learners on a Kaggle telecom dataset using nested cross-validation, PR-AUC, Wilcoxon signed-rank tests, and model-agnostic explanations such as SHAP, partial dependence, and ICE

plots. Their study is particularly relevant because it argues that deployment-ready churn analytics must go beyond a single split and a single accuracy score. El Attar and El-Hajj (2026) extend this deployment logic even further by integrating soft-voting ensembles with SHAP-based explanation, autoencoder-driven customer segmentation, and cost-sensitive threshold selection. Their key contribution is to highlight a recurring gap in the literature: many studies combine only two of the three components of performance, interpretability, and strategic profiling, whereas effective retention systems need all three.

Finally, the reviewed literature reconnects churn analytics with broader business and governance concerns. Omari et al. (2025) explicitly align modeling with KPIs such as CLTV, ARPU, CAC, CRR, and NPS and emphasize transparency and responsible data practices. At the macro level, Imani et al. (2025) identify business-aligned evaluation, adaptive learning, and real-world deployment as unresolved research priorities. These observations are directly relevant to Strategic Information Systems because they suggest that a successful churn project should not stop at model accuracy; it should produce actionable, accountable, and economically meaningful decision support.

Overall, the attached literature suggests four design implications for the proposed project. First, structured telecom churn data should be modeled with strong tabular learners rather than unnecessarily complex architectures. Second, class imbalance and preprocessing must be systematically handled. Third, SHAP- or LIME-based explanations are essential if the system is expected to influence managerial behavior. Fourth, the missing link in much of the literature is the translation of predictions into prioritized customer segments and action rules. The proposed study addresses this gap by designing a churn-centered DSS that combines comparative modeling, explanation, and retention-oriented outputs on a reproducible public dataset.

3. Materials and Methods

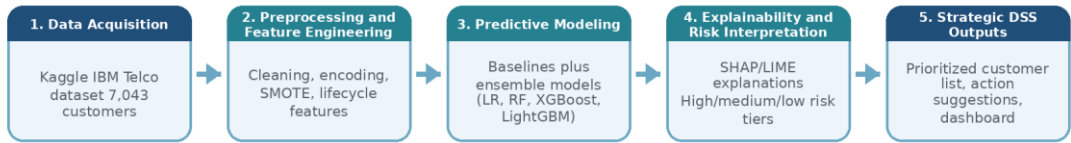
3.1 Research design and system perspective

The project is designed as an artifact-oriented analytical study in which a predictive model is embedded inside a strategic decision support system rather than evaluated in isolation. The planned artifact will consist of five layers: (1) data acquisition, (2) preprocessing and feature engineering, (3) predictive modeling, (4) explanation and risk interpretation, and (5) action recommendation. Its intended users are customer retention teams, marketing analysts, and service managers who require not only a forecast but also a defensible rationale for intervention.

Figure 1 summarizes the proposed workflow. The architecture intentionally connects machine learning outputs to managerial use. In other words, the system is expected to produce customer-level probabilities, feature-level explanations, and action-level suggestions that can support retention campaigns, customer care prioritization, and resource allocation.

Proposed Strategic Decision Support System Workflow

From telecom churn data to interpretable retention actions



Predictive probabilities will be transformed into interpretable, retention-oriented recommendations for managerial users.

Figure 1. Conceptual workflow of the proposed churn-centered strategic decision support system.

3.2 Dataset

The study will use the IBM Telco Customer Churn dataset publicly hosted on Kaggle (Yeanzc, n.d.). This benchmark includes 7,043 customer records and 21 original columns, including the binary target variable Churn and a customer identifier. Recent studies using the same benchmark report a stable class distribution of roughly 73.5% non-churn and 26.5% churn, confirming the presence of moderate class imbalance (Omari et al., 2025; El Attar & El-Hajj, 2026).

Because the dataset is cross-sectional rather than a full event log, it does not support a fully fledged time-series pipeline in the strict sense. Nevertheless, the feature tenure provides a meaningful temporal proxy for lifecycle analysis, and the final study can include a complementary tenure-based or survival-oriented perspective in response to the course recommendation that time-aware methods should be considered where feasible.

Feature group	Examples	Strategic interpretation
Identifier	customerID	Used only for reference and removed before modeling to avoid non-informative leakage.
Demographic profile	gender, SeniorCitizen, Partner, Dependents	Provides customer context and supports aggregate segmentation, but should not be used as an isolated intervention rule.
Account and commitment	tenure, Contract, PaperlessBilling, PaymentMethod	Captures switching barriers, billing behavior, and relationship duration, all of which are central to retention strategy.
Service subscriptions	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	Represents product adoption, perceived service value, and support exposure.
Financial signals	MonthlyCharges, TotalCharges	Reflects revenue contribution, price sensitivity, and cumulative customer value.
Target variable	Churn	Operational outcome to be predicted and transformed into retention-oriented risk categories.

Table 3. Planned analytical use of the IBM Telco dataset variables.

3.3 Planned data preprocessing and feature engineering

All preprocessing will be implemented through leakage-safe pipelines so that transformations are fitted only on training data. This is important because the literature shows that churn results are highly sensitive to data preparation choices, especially when the dataset is moderately imbalanced and contains mixed categorical and numerical variables.

1. Data cleaning: customerID will be removed from the modeling space, duplicate checks will be performed, and blank TotalCharges entries will be converted to missing values.
2. Missing-value treatment: missing TotalCharges values will be imputed with the median, following recent studies that use the same dataset and justify median imputation as a robust choice for skewed billing distributions (Omari et al., 2025; El Attar & El-Hajj, 2026).
3. Encoding and scaling: nominal variables will be one-hot encoded. If exploratory analysis indicates that ordinal representation is useful for commitment-related fields such as contract duration, an alternative encoding will be tested. Numerical variables will be standardized for algorithms that are sensitive to scale, such as Logistic Regression and SVM.
4. Class-imbalance handling: SMOTE will be applied only within the training folds, and class weighting will be used as a robustness check for selected models. This choice is motivated by the literature that consistently identifies imbalance as one of the central technical risks in churn prediction (Imani et al., 2025; Sikri et al., 2024).
5. Feature engineering: in addition to the original variables, the project will test a small set of domain-informed derived variables such as total active services, a month-to-month contract flag, tenure bands, an auto-payment flag, and a support-and-security bundle indicator. These engineered features are intended to improve interpretability as much as predictive power.

A complementary lifecycle-oriented analysis will also be considered by examining churn risk across tenure bands. Although the dataset does not offer full timestamps for event-sequence modeling, this design partially addresses the course recommendation regarding time-aware methods and is consistent with the survival-oriented reasoning used in telecom churn studies such as Wagh et al. (2024).

3.4 Modeling strategy

The predictive core will compare a set of models chosen to balance interpretability and predictive strength. Logistic Regression will function as an interpretable baseline because it allows relatively transparent coefficient-based reasoning. Random Forest will provide a robust nonlinear baseline for structured tabular data. Gradient boosting models such as XGBoost and LightGBM will serve as the primary high-performance candidates, and CatBoost may be added if handling of categorical structure appears advantageous. An RBF-kernel SVM can also be included as a benchmark because IBM Telco studies have found strong results for SVM in this setting (Omari et al., 2025).

The project will not prioritize deep learning as the main modeling strategy in Phase 2 because the selected benchmark is relatively small, highly structured, and frequently dominated by strong tabular learners in the reviewed literature (Imani et al., 2025; Zerine et al., 2026). Instead, if individual models display complementary strengths, the top candidates will be combined in a soft-voting ensemble to improve stability and decision-support reliability.

3.5 Evaluation plan

The dataset will be partitioned using stratified sampling, with 80% reserved for training and 20% for the final hold-out test set. Within the training data, 5-fold stratified cross-validation will be used for hyperparameter tuning and model comparison. If time and implementation complexity allow, a nested cross-validation variant will also be tested as a robustness analysis, following the evaluation logic recommended by Zerine et al. (2026).

Model evaluation will not rely on accuracy alone. The primary performance indicators will be precision, recall, F1-score, ROC-AUC, PR-AUC, and the confusion matrix. PR-AUC is particularly important because the churn class is the minority class and because retention management is sensitive to precision-recall trade-offs. If multiple models perform similarly, a fold-wise statistical comparison such as the Wilcoxon signed-rank test will be considered to support more rigorous model selection.

The classification threshold will not automatically remain at 0.50. Because false negatives are strategically more costly than false positives in retention management, threshold selection will consider both F1-score and a simple cost-sensitive scenario in which missing a true churner is treated as more expensive than contacting a loyal customer. This choice is intended to keep the evaluation aligned with business use rather than purely statistical optimization.

3.6 Explainability and strategic DSS outputs

After the best-performing model is selected, explainability will become the core bridge between analytics and strategic action. SHAP will be used for global feature importance, local customer-level explanations, and the identification of dominant churn patterns across contract type, tenure, service bundle, support status, and billing features. LIME may be used as a secondary local explainer when a case-specific narrative is desirable for stakeholder communication.

Customers will then be categorized into low-, medium-, and high-risk tiers according to predicted probability and threshold analysis. For each medium- and high-risk case, the system will report the churn probability, the main explanatory drivers, and a suggested managerial response such as a contract migration offer, technical support outreach, pricing review, service bundle incentive, or payment-method intervention. In this way, the software implementation planned for Phase 2 will operate as a decision-support artifact rather than a pure prediction engine.

Planned DSS output	Description	Managerial use
Predicted churn probability	Customer-level probability score generated by the selected model.	Prioritize scarce retention resources and define monitoring intensity.
Risk tier (high / medium / low)	Probability transformed into interpretable operational categories.	Support campaign targeting and budget allocation.
Top churn drivers	SHAP/LIME explanation of the strongest customer-specific factors.	Enable personalized intervention and managerial justification.
Segment-level dashboard	Aggregated patterns by contract type, tenure band, payment method, and service bundle.	Reveal structural churn patterns for tactical and strategic planning.

Planned DSS output	Description	Managerial use
Recommended action	Rule-based mapping from risk profile to response alternatives.	Guide customer care, pricing, and loyalty interventions.

Table 4. Planned decision-support outputs of the final system.

3.7 Ethical, practical, and implementation considerations

Although the chosen dataset is public and anonymized, the project will document all preprocessing choices, random seeds, modeling parameters, and evaluation settings to preserve transparency and reproducibility. Demographic variables such as gender and senior-citizen status will be interpreted cautiously and primarily at aggregate level rather than as isolated operational triggers. The DSS is intended to support human decision-makers, not to automate customer treatment without managerial review.

The Phase 2 implementation will be delivered as a working software prototype built around the predictive pipeline described above. Its purpose will be to demonstrate how analytical CRM can be supported by a practical information system that combines predictive probabilities, explanation layers, and retention recommendations in one interface. Therefore, the final contribution of the project is expected to be both technical and strategic: a machine learning solution that is explicitly designed for decision support.

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